
MERVEKS SAP

Enterprise Operations & Finance Console

Client User Guide & System Manual

Version: 1.0 (Production Release)

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Host: <https://merveks-sap.vercel.app>

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1. Executive Summary

The MERVEKS SAP Console is a secure, state-of-the-art Web ERP application designed to manage international trade operations, cargo logistics, warehousing, procurement, and corporate accounting for MERVEKS. From overseeing food supply chains and railway freight routes to coating distribution logistics, this platform serves as a single point of operational visibility.

Built with a focus on ease of use, responsiveness, and speed, the application equips team members with specialized modules tailored to their corporate roles. Every operation - whether recording client requests, issuing sales quotes, tracking cargo loads, logging supplier bills, or balancing cash flows - is fully tracked, permission-gated, and audited.

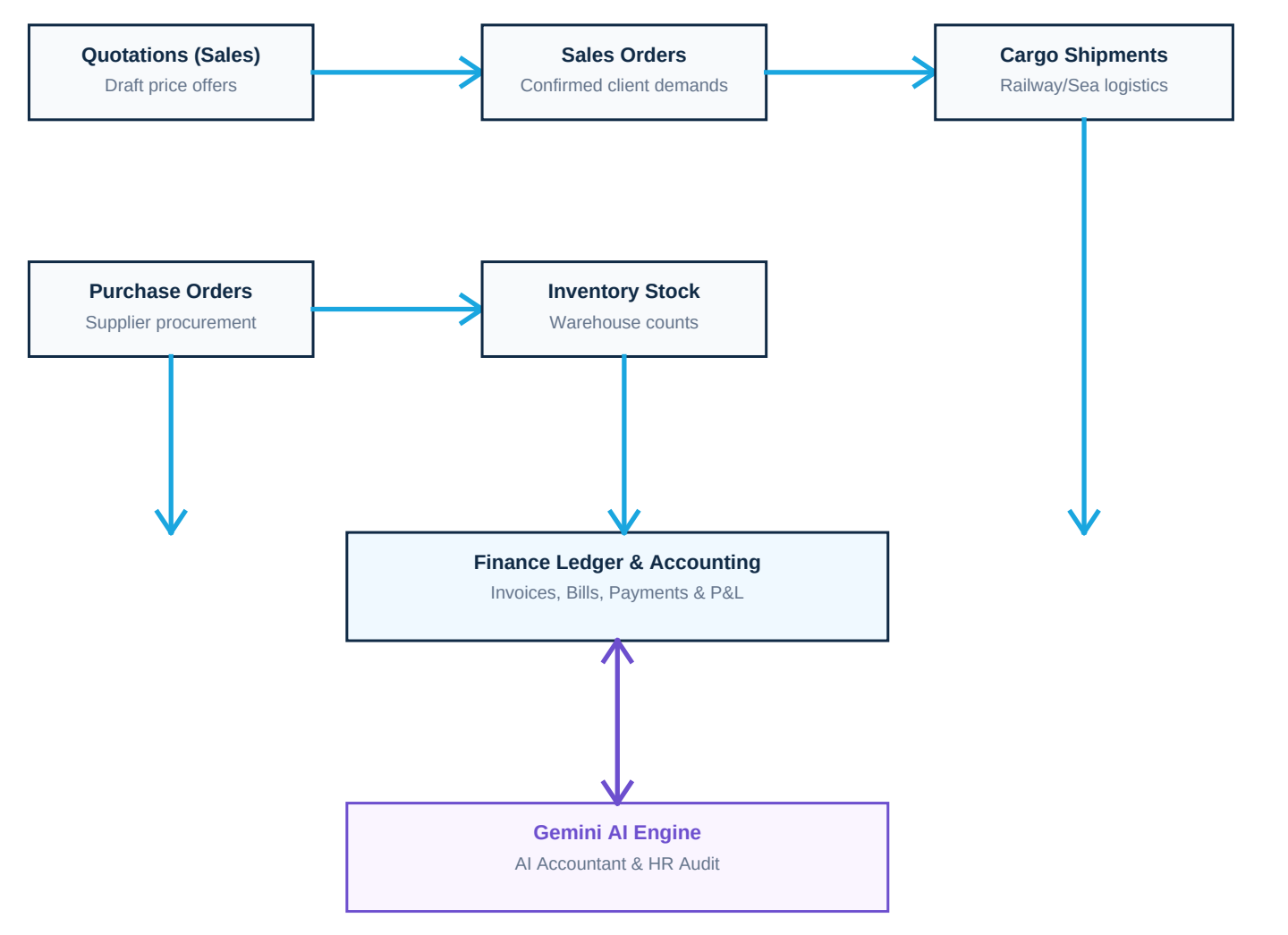
2. System Architecture & Tech Stack

The system utilizes a modern, zero-dependency client-side architecture to deliver high speed and maximum reliability:

- Frontend Shell: Standard HTML5 and Vanilla CSS variables with custom Jost & Inter typography.
- Application Logic: Component-driven modular ES6 Javascript. It follows a clean IIFE architecture that eliminates dependencies on heavy web frameworks, ensuring sub-second load times.
- Data & Connection Layer: Connected to local indexed storage for offline performance and configured to synchronize with the MERVEKS ERP REST APIs. Disconnecting enters the mock sandbox mode.
- Security: Client-side routing checks and strict role boundary validations at the component level.
- Deployment: Hosted and built automatically via Vercel Edge Server infrastructure.

3. System Operational Workflow

The console coordinates sales, inventory, logistics, and finance into a single unified web portal. The flowchart below illustrates how these entities connect and work together dynamically:



Note: Every database modification logs audit records in the Activity History ledger. Stock values recalculate dynamically when POs are received (replenishing warehouse) or cargo is dispatched. The AI accountant processes the live ledger to flag risk and credit.

4. Sales & Order Management (Client Pipeline)

The sales module is structured around Quotations and Orders, reducing manual steps:

- Quotations: Trade specialists can draft detailed quotes for prospects, selecting the client, currency (USD, EUR, TRY), payment terms, and adding multiple line items with specific unit prices. Approved quotes transition to orders with a single click on 'Accept & Create Order'.
- Sales Orders: Tracks active orders through confirmation, shipping, completion, or cancellation. Order pages gate access dynamically, preventing unauthorized modifications.

5. Logistics & Cargo Tracking (QR Labeling)

Logistics tracking is designed for high-resolution tracking of international cargo:

- Core Metrics: Tracks origin, destination, transport mode (Railway, Sea, Land), weight in tons, and container quantities.
- Route Timeline: Operators record tracking check-ins (checkpoints) to chart transit progress.
- QR Labeling: Clicking 'Print Label' opens a layout containing a system-generated QR code. Field workers can print this label and scan it with a mobile camera via the built-in scanner to update checkpoints.
- Public Tracking Portal: Clients scan the QR code to open the public tracking portal. They see routes, container counts, departed dates, ETA, and live timelines without needing login credentials.

6. Procurement & Inventory Control

Inventory levels and supply lines are kept in balance via integrated PO operations:

- Purchase Orders (PO): Purchase orders are logged to suppliers to replenish warehouse stock. When a PO is received physically, clicking 'Receive & Create Bill' automatically increments stock levels for the related product and creates an unpaid supplier bill in the accounting ledger.
- Low Stock Warnings: The inventory system constantly monitors product counts. If warehouse quantities fall below the pre-configured reorder threshold, visual warnings like 'Low Stock' are triggered, prompting trade managers to issue purchase orders.
- Warehousing Facilities: Tracks storage locations (e.g. Mersin Main, Istanbul DC) and displays total estimated inventory value in real-time.

7. Finance, Accounting & AI Accountant

The system offers advanced finance ledgers integrated with AI automation:

- Finance Ledger: Tracks active Receivables (AR) and Payables (AP). Payments recorded in the system automatically recalculate the net cash flow and balance sheet figures.
- Aging Receivables: Computes outstanding customer debt and groups it into chronological periods (Current, 1-30 days, 31-60 days, 60+ days) represented by visual progress bars.
- PDF & Excel Sheets: Generates detailed statements and individual invoice documents with a single click. Row actions inside the ledger allow printing separate invoice sheets, and 'Export Excel' downloads formatted CSV data.
- Gemini AI Accountant: Powered by Google Gemini. Evaluates open invoices, reviews payment history, and drafts a credit-risk report. It identifies overdue clients, analyzes collection speed, and suggests mitigation strategies.
- AI Performance & Raise Recommendations: The system computes employee rankings using multivariable metrics (Quality, Consistency, Revenue impact, and Seniority). Based on these metrics, the AI suggests annual raise options (e.g. +8% to +11%) or flags a Performance Improvement Plan (PIP) if a user requires support.

8. Role-Based Access Controls (RBAC) Details

Strict permission boundaries protect corporate databases and govern system views:

- Owner: Has absolute system permission. Only owners can access margins, see complete audit timelines, customize employee account stats, log salary raises, or delete database elements.
- Operations Manager: Scoped to transport, logistics, inventory stock, and orders. Sidebar automatically filters out AR/AP tabs, salary details, profit & loss, and margins. Direct URL route tampering attempts are blocked by the router and redirect managers back to the dashboard.
- Finance Officer: Scoped to accounting ledger, client billing, payments, and AI auditing. Sidebar only lists Dashboard, Clients, and Finance. Direct operations tabs are gated and completely locked.
- Trade Specialist: Scoped to quotes, orders, and clients. Action buttons to delete entries or modify team parameters are hidden.

9. Interface Customizations & Settings

The interface incorporates premium design choices to elevate user experience:

- Collapsible Sidebar: Click 'MENU' next to the three bars to slide the sidebar off-screen, expanding the workspace. The expanded/collapsed state is saved to browser localStorage.
- Custom Language Dropdown: Placed in the top right. Allows changing interface language to Türkçe, English, or Rusça. Uses the signature merveks.com design (bold red border, sharp corners, and blue selection state).
- User Profile Page: Accessible by clicking the top right user menu card. Allows users to view their role details, performance scores, salary history, raise timeline, and personal activity log.

10. Access Instructions & Quickstart

To access the console and start testing the application:

1. Open the URL: <https://merveks-sap.vercel.app>
2. Log in using the demo credentials provided on the login page:
 - Owner: owner@merveks.com (Password: merveks2013)
 - Operations: operations@merveks.com (Password: ops123)
 - Finance: finance@merveks.com (Password: fin123)
3. Switch languages at the top right to verify localization overrides.
4. Collapse the sidebar using the MENU button to verify fullscreen transitions.
5. Go to Accounting to test the AI Accountant Gemini analysis or export PDF/Excel statements.
6. Click your user card at the top right to inspect your personal employee profile.

11. System Interface Screenshots

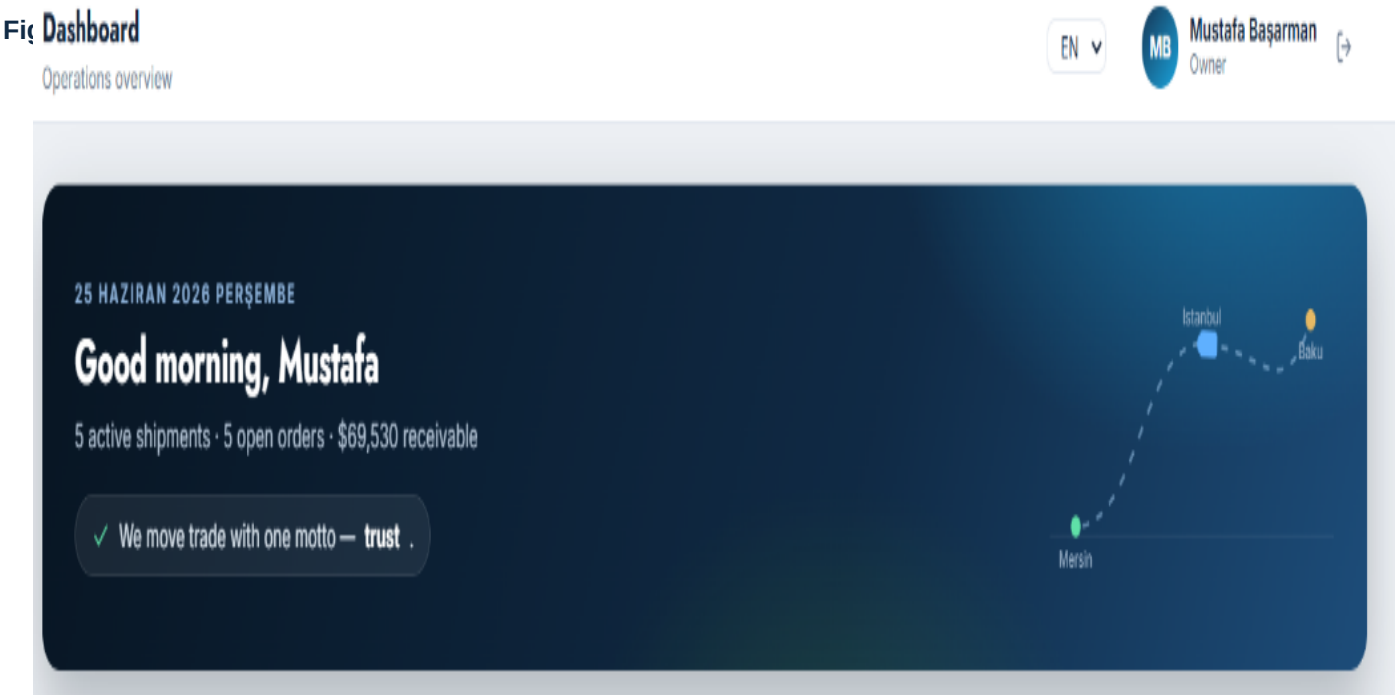


Figure 2: Quotations & Sales Orders Pipeline Management

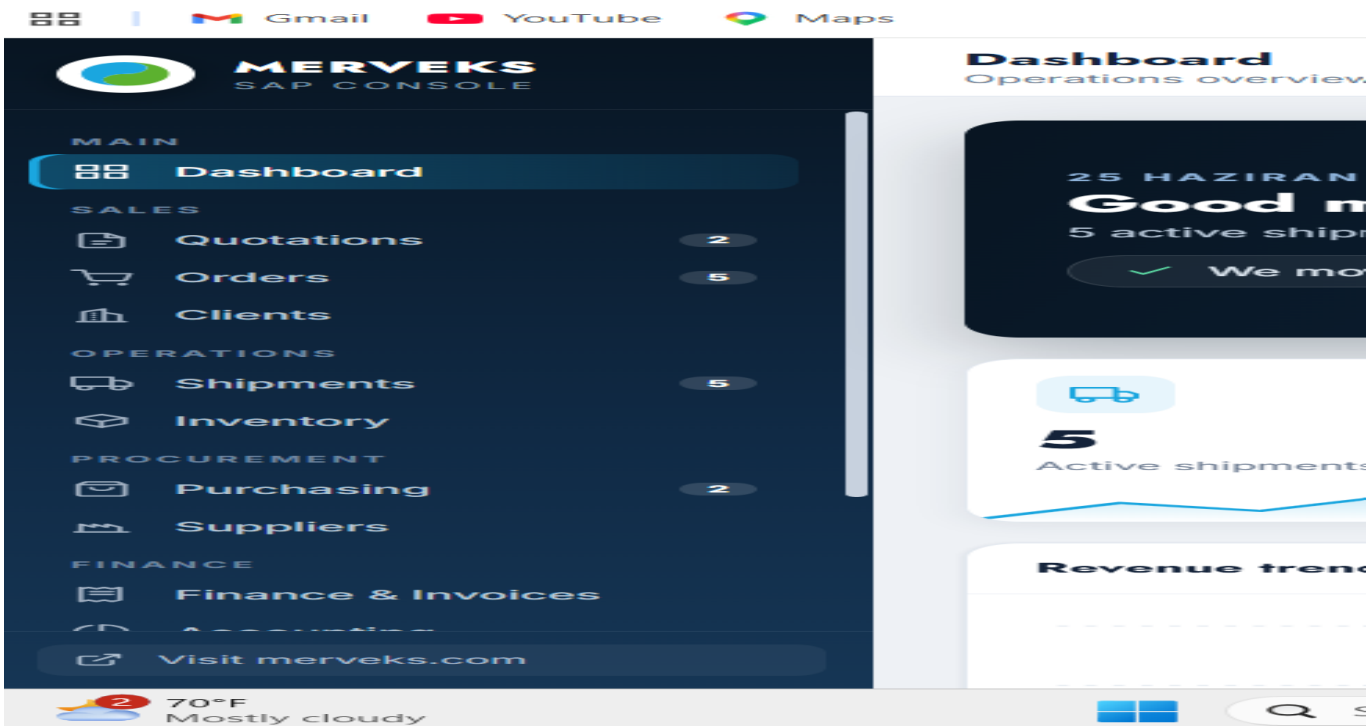




Figure 4: Detailed Employee Performance Profile Dashboard

